ICSE

ANALYSIS OF PUPIL PERFORMANCE

October 2025

COMPUTER APPLICATIONS

Research Development and Curriculum Division
Council for the Indian School Certificate Examinations
New Delhi

OCTOBER 2025

© Copyright, Council for the Indian School Certificate Examinations

All rights reserved. The copyright to this publication and any part thereof solely vests in the Council for the Indian School Certificate Examinations. This publication and no part thereof may be reproduced, transmitted, distributed or stored in any manner whatsoever, without the prior written approval of the Council for the Indian School Certificate Examinations.

Council for the Indian School Certificate Examinations (CISCE)

MISSION STATEMENT

The Council for the Indian School Certificate Examinations is committed to serving the nation's children, through high quality educational endeavours, empowering them to contribute towards a humane, just and pluralistic society, promoting introspective living, by creating exciting learning opportunities, with a commitment to excellence.

ETHOS OF CISCE

- Trust and fair play.
- Minimum monitoring.
- Allowing schools to evolve their own niche.
- Catering to the needs of the children.
- Giving freedom to experiment with new ideas and practices.
- Diversity and plurality - the basic strength for evolution of ideas.
- Schools to motivate pupils towards the cultivation of:
  Excellence - The Indian and Global experience.
  Values - Spiritual and cultural - to be the bedrock of the educational experience.
- Schools to have an 'Indian Ethos', strong roots in the national psyche and be sensitive to national aspirations.

FOREWORD

The Council for the Indian School Certificate Examination has endeavoured to sincerely take forward the vision of NEP 2020 in enhancing pedagogy and assessments through systemic reforms. In alignment with this vision, the CISCE also remains committed to ensuring that learning is meaningful, and that assessment practices are utilised as tools for growth, reflection, and continuous improvement rather than as an exercise in mere evaluation.

The ISC and ICSE examinations have long served as the cornerstone of the CISCE's academic framework, especially in shaping the educational journeys and futures of countless CISCE students. In this context, the *Analysis of Pupils' Performance* documents have played a significant role in strengthening the CISCE's assessment culture. These reports have, over the years, supported students in understanding subject-wise expectations and in preparing effectively for the Board examinations.

Our heartfelt appreciation is extended to the Research Development and Curriculum Division (RDCD) of the CISCE for their dedicated efforts in preparing this detailed document for the benefit of students. We also express our gratitude to the examiners for their meticulous evaluation and insightful feedback on candidates' performance, as well as for recommending pedagogical strategies to help teachers address common areas of difficulty.

We are confident that students, teachers, and parents will make the best use of this document by engaging with its various sections in depth and by implementing the learnings to ensure enhanced and successful performance in the forthcoming examinations.

October 2025

Dr. Joseph Emmanuel
Chief Executive & Secretary
CISCE

PREFACE

The *Analysis of Pupil Performance* document is a standing legacy of the Council for Indian School Certificate Examinations (CISCE) that aims at capturing performance trends of CISCE candidates in the annually conducted ICSE and ISC examinations. Over the years, these user-friendly and accessible reports have helped CISCE schools analyse patterns in candidates' responses to board examination papers, while also providing valuable pedagogical insights and suggestions for teachers.

Each year, CISCE undertakes this exercise of a detailed analysis of the performance of candidates for selected subjects at both the ICSE and ISC examinations. Each subject document includes the questions from the board examination along with the marking scheme for each question, enabling teachers and students to better understand the scope of the question and the appropriate approach to answering it. It also presents an infographic presentation of statistical analysis and contextualised commentaries on candidate performance with detailed insights and remedies on the errors. Towards the end of the document, a comprehensive summary highlights the topics revealing gaps in conceptual understanding, areas that candidates found confusing, and tailored suggestions to help students approach examinations more effectively.

The uniqueness of this document lies in its micro-analysis of items in board examination papers where suggested pedagogical strategies are tied to each common errors identified from candidates' answers, a strategy that CISCE believes is rudimentary in driving improvements rather than merely sharing a broad stroke macro analysis of candidate performance. While these reports are grounded in real candidate data, they also lend themselves to seamlessly to classroom action.

The Analysis of Pupil Performance document for ICSE for the Examination Year 2025 covers the following subjects - English Language, Literature in English, Hindi, History and Civics, Geography, Mathematics, Physics, Chemistry, Biology, Commercial Studies, Economics, Computer Applications, Economic Applications, Commercial Applications, Environmental Science and Home Science.

The subjects covered in the ISC Analysis of Pupil Performance document for the Year 2025 are - Accounts, English Language, Literature in English, Hindi, Economics, Commerce, Business Studies, Mathematics, Physics, Chemistry, Biology, Elective English, History, Political Science, Geography, Psychology, Sociology, Computer Science, Environmental Science, Home Science and Legal Studies.

I extend my heartfelt gratitude to all the ICSE and ISC examiners who have shared their valuable comments on each question. I also acknowledge the efforts of the RDCD team of Dr. Manika Sharma, Ms. Parul Kohli, Ms. Lyimee Saikia, Ms. Mansi Guleria, Ms. Gunjan Khurana, Ms. Madhusree Chatterjee and Ms. Aakriti Agrawal for their commitment to ensuring the compilation of the comments and data into its current form.

We hope that the document is successful in its intended mission of better assisting our teachers in identifying areas that require focused intervention and allowing them to better support CISCE students. At the same time, we also hope that it guides our students in developing a roadmap of effective study methods for their examinations.

October 2025

Dr. Bhawna Taragi
Deputy Head
RDCD, CISCE

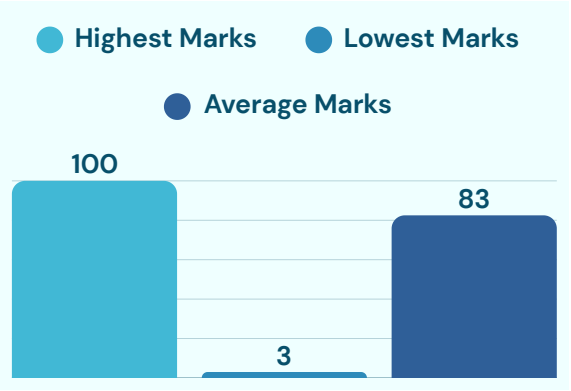
QUANTITATIVE ANALYSIS

Computer Applications

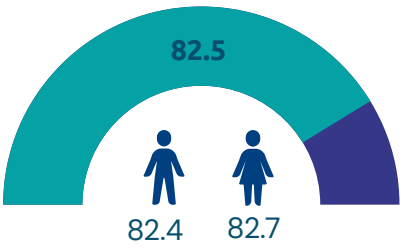
Number of Candidates Appeared
143269

Boys	Girls
79467	63802

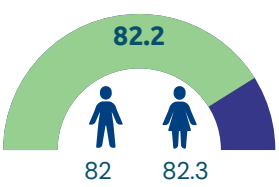
Maximum Marks **100**



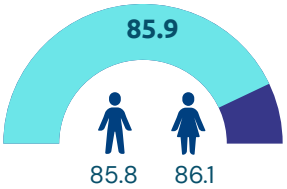
Overall Performance (Mean Marks)



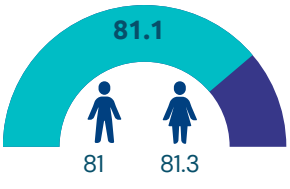
Region-wise Performance (Mean Marks)



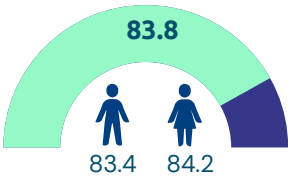
Eastern



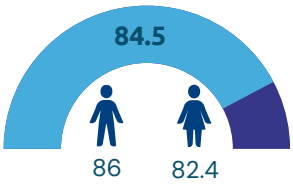
Western



Northern



Southern



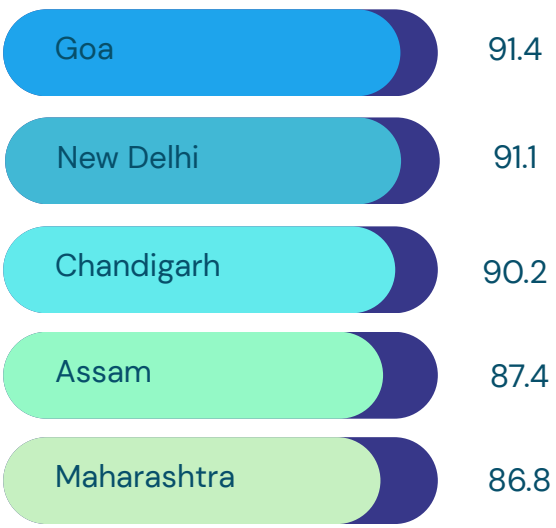
Foreign

Performance by Gender

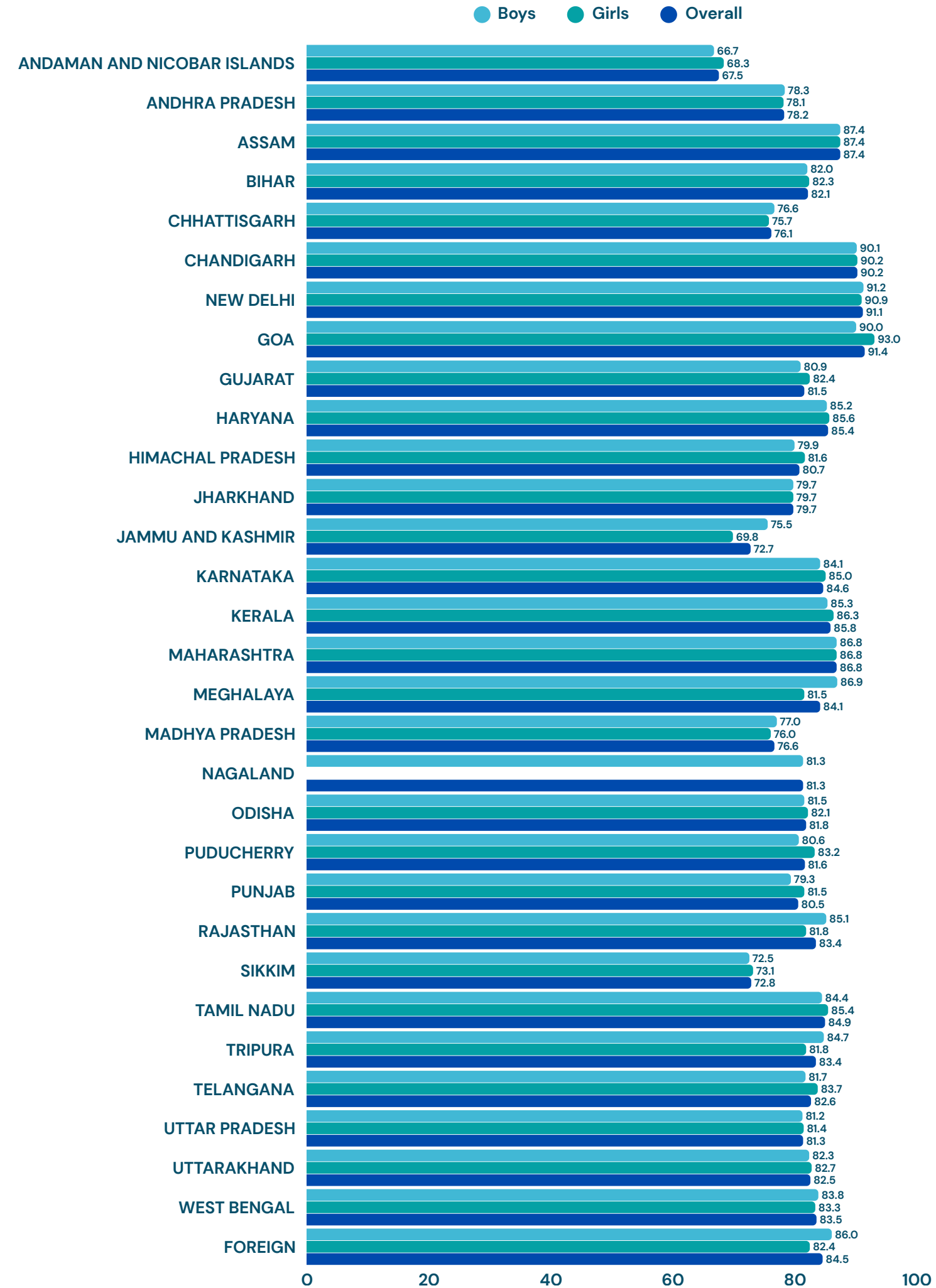
Gender	Mean	SE	t-value
Girls	82.7	0.05	4.2*
Boys	82.4	0.05	

*Significant at 0.05 level

Top 5 Perfoming States/UTs (Mean Marks)



Performance State-wise and Foreign (Mean Marks)



SECTION A (40 MARKS)

(Attempt all questions from this Section.)

Question 1

[20]

Choose the correct answers to the questions from the given options.

(Do not copy the questions, write only the correct answers.)

(i) Character class methods are found in the ***package*** called:

(a) java.util

(b) java.lang

(c) java.awt

(d) java.io

(ii) System.out.println**('Z'+32)**; will display:

(a) z

(b) Z

(c) 122

(d) 154

(iii) ***double x [ ]={2.5,4.5,5.5,6.4};*** occupies _____ bytes.

(a) 16

(b) 4

(c) 8

(d) 32

(iv) The output of ***42/6%2*** is:

(a) 1

(b) 10

(c) 2

(d) 0

(v)

 **Mouse**	 **Keyboard**	 **Printer**

 **Scanner**	 **Barcode Scanner**	 **Microphone**

Consider the ***Two dimensional array P[2][3]***, of peripherals (input / output devices) given above, state the index of the device ***Barcode Scanner***.

(a) P[1][1]

(b) P[0][1]

(c) P[1][2]

(d) P[0][0]

(vi) Which of the following is ***user defined data type***?

1. array	3. class
2. double	4. boolean

(a) only 1

(b) 1 and 3

(c) only 2

(d) only 4

(vii) Select the ***infinite*** loop:

(a) for(int i=1;i<=10;i++)

(b) for(int i=2; i!=0 ;i-=3)

(c) for(int i=5 ;i<=5;i++)

(d) for(int i =1;i>=1; i - -)

(viii) The output of ***Math.max(-7, Math.min(-5 ,-4))*** is:

(a) –5

(b) –4

(c) –7

(d) error

(ix) Which of the following ***is true*** for the given object creation statement?

Game cricket = new Game( );

(a) Game is an object of cricket class

(b) New keyword creates object Game

(c) Game is a class and cricket is an object

(d) Game and cricket are objects

(x) ***Post office*** is an example for _____ access specifier.

(a) public

(b) local

(c) protected

(d) private

(xi) **Assertion (A):** In switch case ***break*** statement avoids fall through.

Reason (R): ***break*** statement helps to execute only one case at a time.

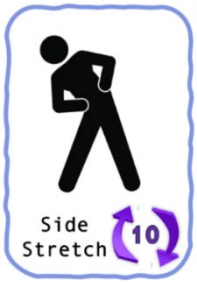
(a) Both (A) and (R) are true and (R) is a correct explanation of (A).

(b) Both (A) and (R) are true and (R) is not a correct explanation of (A).

(c) (A) is true and (R) is false.

(d) (A) is false and (R) is true.

(xii) A physical education teacher asks the students to do the ***side stretch*** as shown below,10 times. Which ***programming construct*** the teacher uses?



(a) if

(b) switch

(c) for

(d) if else if

(xiii) The ***index (subscript)*** of the last element of an array ***ar[ ]*** is:

(a) ar.length( )

(b) ar[ ].length

(c) ar.length()-1

(d) ar.length-1

(xiv) **Assertion (A):** A clock is a real-life example of ***nested loops***.

Reason (R): The hour hand moves through 12 positions, while the minute hand moves through 60 positions within each hour.

(a) Both (A) and (R) are true and (R) is a correct explanation of (A).

(b) Both (A) and (R) are true and (R) is not a correct explanation of (A).

(c) (A) is true and (R) is false.

(d) (A) is false and (R) is true.

(xv) Which of the following ***pairs of methods*** will cause a ***compile-time error*** due to incorrect method ***overloading***?

(a) void test(int a, int b) and void test(double a, double b)

(b) void test(int a, double b) and void test(double a, int b)

(c) void test(int a, double b) and void test(int a)

(d) void test(int a) and int test(int a)

(xvi) Which of the following converts ***"25"*** to ***25.0***?

(a) Double.Parsedouble("25")

(b) Double.parse("25")

(c) Double.parseDouble("25")

(d) Double.parseDouble(25)

(xvii) Consider the program segment:

```
int p=0;
for(p=4 ; p>0 ; p- =2);
System.out.print(p);
System.out.println(p);
```

The above statements will display:

(a) 42

(b) 4200

(c) 0
0

(d) 00

(xviii) ***System.out.println("I said,\"It\'s wise to obey elders.\"");***

The ***output*** of the above statement is:

(a) I said,'It is wise to obey elders.'

(b) I said, "It's wise to obey elders."

(c) I said,It's wise to elders.

(d) "'It's wise to obey elders.'"

(xix) What is the ***output*** of the statement given below?

"ANGER". compareTo("ANGEL")

(a) 3

(b) –6

(c) 6

(d) 0

xx) Consider the following program segment in which the statements are ***jumbled***. Choose the ***correct order*** of statements to calculate and return the ***factorial of 4***.

for (k=1; k<=4; k++) ⟶ 1

return fa; ⟶ 2

long fa = 1, k; ⟶ 3

fa*=k; ⟶ 4

(a) 1, 2, 3, 4

(b) 3, 1, 4, 2

(c) 3, 1, 2, 4

(d) 1, 3, 2, 4

MARKING SCHEME

(i)	(b) java.lang
(ii)	(c) 122
(iii)	(d) 32
(iv)	(a) 1
(v)	(a) P[1][1]
(vi)	(b) 1 and 3
(vii)	(b) for(int i=2; i!=0 ;i-=3)

(viii)	(a)	–5 or (d) error
(ix)	(c)	Game is a class and cricket is an object
(x)	(a)	public
(xi)	(a)	Both (A) and (R) are true and (R) is a correct explanation of (A).
(xii)	(c)	for
(xiii)	(d)	ar.length-1
(xiv)	(a)	Both (A) and (R) are true and (R) is a correct explanation of (A).
(xv)	(d)	void test(int a) and int test(int a)
(xvi)	(c)	Double.parseDouble(“25”)
(xvii)	(d)	00
(xviii)	(b)	I said, “It's wise to obey elders.”
(xix)	(c)	6
(xx)	(b)	3, 1, 4, 2

Comments of Examiners

(i) Most of the candidates chose the correct option (b) i.e. java.lang package. A few candidates incorrectly chose the option (a).

(ii) A few candidates got confused with the concept of implicit typecasting, thus choosing the incorrect options (a) or (d).

(iii) Most of the candidates answered it correctly. However, a few candidates chose the wrong options.

(iv) Majority of the candidates were able to give the correct option. A few candidates opted for option (d), which was incorrect.

(v) Many candidates could answer this question well.

(vi) Most candidates chose the correct option (b). A few candidates opted for option (a) and lost marks.

(vii) Few candidates got confused as all the loop statements looked alike, and thus chose the wrong options.

(viii) Most of the candidates answered this part correctly.

(ix) A few candidates chose the wrong option.

(x) Majority of the candidates answered it correctly. However, a few of them gave the incorrect option (c).

(xi) Most of the candidates chose the correct option.

(xii) This question was answered correctly by most of the candidates.

(xiii) Few candidates got confused between options (c) and (d).

(xiv) Majority of the candidates answered this question correctly.

(xv) This question was answered correctly by many of the candidates.

(xvi) Most of the candidates got this question right. Some candidates also chose the incorrect options.

(xvii) Many candidates did not consider the semicolon at the end of the 'for statement' and also ignored the System.out,print() followed by System.out.println(), resulting in the incorrect answer.

(xviii) A few candidates got confused with the punctuation marks given in the question and thus wrote the incorrect answer.

Suggestions for teachers

- Teach the concept of packages java.lang, java. util, default package java.lang and the need for an import statement for the utility package.
- Provide sufficient practice of multiple-choice questions.
- Explain the concept of implicit and explicit type casting with a variety of output-based questions with different data types for better conceptual clarity.
- Explain the definition of arrays, how to declare the arrays, the number of elements in the array, the size in bytes and bits occupied by the array of different primitive datatypes.
- Ensure thorough practice on the evaluation of a variety of mathematical expressions involving different combinations of the arithmetic operators.
- Provide more practice for evaluation of the expression from left to right of the expression following the hierarchy of the operators and not the BEDMAS or BODMAS rules.
- Guide the students that the index begins from 0.
- Explain explicitly the concept of primitive datatypes and user-defined data types / non primitive data types.
- Teach the concept of types of loops by giving examples.
- Ensure that the students do a dry run to check for the type of the loop [finite or infinite].
- Provide thorough exercises on Iterative statements.
- Give thorough practice of Math method outputs with the help of short, simple exercises in class as a quiz or as a game.
- Clearly explain the concept of creating an object of the class by providing a simple syntax as -<class name><object name > = new <class name>();
- Explain the concept of encapsulation and the visibility rules with the help of examples.
- Ensure adequate practice of Assertion and Reason type questions for all the concepts.
- Highlight the difference between length(number of elements in an array) and length() [method which results in the number of characters in a string].
- Teach the concept of the wrapper class methods with appropriate syntax and the set of arguments.

(xix) Most of the candidates answered this question correctly.

(xx) This question was answered well by most of the candidates.

Suggestions for teachers

- Lay emphasis on the effect of a semicolon at the end of a for loop along with the importance of println( ) and print( ) by giving output questions.
- Clearly highlight the effect of print and println in displaying the results.
- Emphasise on the importance of [\] [back slash) while teaching the escape sequences by providing a number of examples.

Question 2

(i) Write the ***java expression*** to find the product of square root of **P** and the square root of **Q** using the methods of ***Math class***. **[2]**

(ii) Write the ***output*** of the following String method: **[2]**

```
String x= " talent" ;  String y="matrix" ;
System.out.print(x.substring(3).concat(y.substring(3))) ;
```

(iii) Write the Java statement for creating an object named ***'sifra'*** of the class ***'Robot'***, which takes ***three double parameters***. **[2]**

(iv) Convert the given loop into **exit controlled loop.** **[2]**

```
int a,b;
for ( a=10 ,b=1; a>=1 ;a-=2)
{
b+=a;
b++;
}
System.out.print(b);
```

(v) Consider and give the ***output*** of the following program: **[2]**

```
class report
{   int a,b;
  report()
  {  a=10;
     b=15;
  }
  report(int x, int y)
  {   a=x;
     b=y;
  }
```

```
    void print()
    {   System.out.println(a*b);
    }
    static void main()
    {   report r = new report();
        r.print();
        report p = new report(4, 5);
        p.print();
    }
}
```

(vi) (a) Name one ***String method*** which results in ***positive integer only***. **[2]**

(b) Name one ***String method*** which results in a ***character***.

(vii) John was asked to write a Java code to calculate the ***surface area of a cone***, the following code was written by him: **[2]**

Surface area of cone is A=πrl $l = \sqrt{r^2 + h^2}$

```
class area
{     double area (double r, double h)
   {  double l, a;
      a=22.0/7*r*l;
      l=Math.sqrt(r*r+h*h);
      return a;
   }
}
```

Specify the ***type of the error*** in the above program, correct and write the program to be error free.

(viii) Consider the following array and answer the questions given below: **[2]**

int a[ ] = {12,10,8,4,6,2,3,5,7}

(a) What is the ***output*** of System.out.print***(a[0]+a[5])***;?

(b) What is the ***index (subscript)*** of the ***largest*** element of the array ***a[ ]***?

(ix) (a) Write the Java statement to initialise the ***first 6 odd numbers*** in a ***3 × 2*** array. **[2]**

(b) What is the result of ***x[0][1] + x[2][1]*** of the above array?

(x) Give the ***output*** of the following program segment and specify ***how many times*** the loop is executed. **[2]**

String s = "JAVA";

```
for(i=0;i<s.length();i+=2)
System.out.println(s.substring(i));
```

MARKING SCHEME	
(i)	Math.sqrt(P) * Math.sqrt(Q) / Math.Pow(P,1.0/2) * Math. Pow(Q,1.0/2) / Math.sqrt(P *Q) / Math. Pow (P *Q ,1.0/2)
(ii)	entrix/ lentrix
(iii)	Robot sifra = new Robot(double a, double b, double c); sifra Robot = new sifra(); Robot sifra = new Robot (a, b, c) Robot sifra = new Robot (any three values) [int/float/char/double

(iv)	int a=10 , b=1;
	do
	{
	b+=a;
	b++;
	a-=2;
	}while(a>=1);
	System.out.print(b);

(v)	(a) 150 (b) 20
(vi)	(a) length ( ) (b) charAt(int)

(vii)	Syntax error / Logical / Compile time error
	double l, a;
	l=Math.sqrt(r*r+h*h);
	a=22.0/7*r*l;
	return a;

(viii)	(a) 14 (b) 0 / a[0]
(ix)	(a) int *x*[ ] [ ] = {{1, 3}, {5, 7}, {9, 11}}; (b) 3 + 11 = 14
(x)	JAVA VA Two times

Comments of Examiners

(i) Most of the candidates were able to convert the statement into a Java expression. However, a few candidates missed out the * symbol.

(ii) This question had a missing bracket; thus, marks were allotted if a candidate attempted this question. Many of the candidates answered this question correctly.

(iii) Most of the candidates gave the correct answer; however, a few missed out on the argument part.

(iv) Many candidates could not attempt this full question correctly as they were not clear with the concept of exit controlled statement, and also missed the semicolon at the end of the while statement. Some candidates missed either the updating statement or the multiple initialisation, due to which they scored only one mark.

(v) Some candidates wrote the answer in reverse order. Very few candidates got this question correct.

(vi) (a) Most of the of candidates were not able to answer this part of the question correctly. Several candidates wrote compareTo, index of (), lastIndexOf() and lost marks.

(b)Most of the candidates wrote the correct answer as charAt(). A few candidates wrote the 'Character class methods' as they did not read the question carefully, which mentioned 'String method'.

(vii) Most of the candidates wrote the incorrect answer as 'logical error'.

(viii) (a) Most of the candidates wrote it correctly.

(b) Many candidates could answer this part correctly. However, some of the candidates wrote the answer as a [0] for which a mark was allotted.

(ix) (a) Many candidates wrote in matrix form. Some candidates wrote Initialisation of Double Dimensional Array incorrectly, also random even / odd numbers with no curly brackets were given.

(b) Based on the values given in ix(a), the candidates answered, and accordingly, marks were allotted.

(x) Marks were given for this question as the bracket was missing in the program segment.

Suggestions for teachers

- Clearly explain java expressions with math method and give clear instructions with respect to the parenthesis and operators.
- Provide adequate practice by giving more expressions.
- Provide practice on output statements involving multiple string methods.
- Explain the statement to create an object of a class with proper syntax.
- Teach the concept of entry-controlled loops [for and while] and the exit controlled loop [do while] by explaining the syntax and working in detail followed by conversion of one iterative statement to another.
- Instruct the students to read the question carefully.
- Give a variety of loop structures for conversion.
- Ensure that students are clear with the concept of initialization, process to be performed, and condition to be checked when converted to while and do while.
- Explain the concept of constructor overloading by solving suitable programs.
- Ensure that the students understand the concept of creating multiple objects invoking multiple constructors following the concept of overloading.
- Execute such programs and let the students understand which constructor is invoked based on the arguments given while creating the objects.
- Provide a code with the constructors having same parameter list to the students and make them identify the errors in it.
- Explain string methods specifying the name of the method, purpose of the method, syntax, arguments if any to be given, return type of the method followed by ample number of examples.
- Instruct students to answer the questions on methods(char/Math/String) in an appropriate manner.
- Clarify the flow of statements/calculations by giving examples.
- Explain the concept of initialization of double dimensional array based on the specifications [order] by giving variety of examples and exercises.

SECTION B (60 MARKS)

*(Attempt **any four** questions from this **Section**.)*

Question 3

[15]

Define a ***class*** named ***CloudStorage*** with the following specifications:

- ***Member Variables:***
  - int acno - stores the user's account number.
  - int space - stores the amount of storage space in GB purchased by the user.
  - double bill - stores the total price to be paid by the user.
- ***Member Methods:***
  - void accept() - prompts the user to input their account number and storage space using Scanner class methods only.
  - void calculate() - calculates the bill total price based on the storage space purchased using the pricing table provided:

Storage range	Price per GB (Rs)
First 15 GB	15
Next 15 GB	13
Above 30 GB	11

  - void display() - displays the account number, storage space and bill to be paid.

Write a ***main method*** to create an ***object*** of the class and ***invoke*** the methods of the class with respect to the object.

MARKING SCHEME

```
import java.util.*;
class CloudStorage
{
   int acno,space;double bill;
   Scanner ob = new Scanner (System.in);
   void accept()
   {
      System.out.println("Enter Account number");
      acno=ob.nextInt();
      System.out.println("Enter Storage Space");
      space=ob.nextInt();
   }
   void calculate()
   {
      if(space<=15)
      bill=space*15;
      else
      if(space>15&&space<=30)
      bill=15*15+(space-15)*13;
      else
      bill=15*15+15*13+(space-30)*11;
   }
   void display()
   {
      System.out.println("Account Number="+acno);
      System.out.println("Storage Space="+space);
      System.out.println("Bill to be paid="+bill);
   }
   void main()
   {
     CloudStorage c = new CloudStorage();
     c.accept();
     c.calculate();
     c.display();
   }
   }
```

Comments of Examiners

Some common errors made by the candidates were:

- Majority of the candidates did not use the slab/tier method to calculate the bill.
- Many of them were unaware of creating an object and call the methods too.
- Some of the candidates accepted the value of the bill.
- Some candidates used the incorrect class name, member variables.
- Some of the candidates declared the member variables inside the accept( ) method.
- Incorrect declaration of scanner class object and the methods to accept the respective values.
- Calculations were not as per the specifications given.
- Some candidates initialized the data member as bill and stored the bill amount in another new variable, which was not a data member.
- display( ) – only the bill was printed.
- A few candidates did not declare the member variable bill.

Suggestions for teachers

- Instruct the students to follow the variables names, method names as given in the question.
- Encourage students to follow step wise representation of the program as per the specifications given.
- Provide sufficient practice of these questions specifying how to represent systematically declare variables, create an object of the scanner class, and accept the required values using scanner class methods in the accept ( ) method.
- Elaborate on how to calculate the required value as per the specifications in the appropriate method, display the contents, create the object in the main method, and invoke the methods with respect to the object created.

Question 4

[15]

Define a ***class*** to accept values into a 4 x 4 integer array. Calculate and print the ***NORM*** of the array.

NORM is the square root of sum of squares of all elements.

1	2	1	3
5	2	1	6
3	6	1	2
3	4	6	3

Sum of squares of elements = 1+4+1+9+25+4+1+36+9+36+1+4+9+16+36+9 = 201

NORM= Square root of 201 = 14.177446878757825

MARKING SCHEME

```
import java.util.*;
class norm
{
   int x[][]=new int[4][4];
   Scanner ob = new Scanner (System.in);
   void calculate()
   {
      int r,c,sum=0;
      //accept the values in the array
      for(r=0;r<4;r++)
      {
         for(c=0;c<4;c++)
         {
           System.out.println("Enter value");
            x[r][c]=ob.nextInt();
            sum+=x[r][c]*x[r][c];
         }
      }
      System.out.println("Norm="+Math.sqrt(sum));
   }
}
```

Comments of Examiners

Some common mistakes made by the candidates were:

- Declaration and accepting the values in a double dimensional array.
- Sum not initialized to zero.
- Most of the candidates forgot to sum up the squares of each element.
- Square root of the sum was ignored.
- The nested for loop was done incorrectly. Loops were from 1 to 4 instead of 0 to <4.
- Missed subscripts while inputting.
- a[i][j] was written as a[ ][ ] =sc.nextInt();.
- Wrote Math.pow(s,2) instead of Math.sqrt(s).

Suggestions for teachers

- Guide the students to analyse the question in order to code the program that is input -process-output.
- Teach the concept of 2D array by encouraging the students to declare the array as per the specifications [order of the array], accept the values into the array using Scanner class methods in an appropriate nested for loop.
- Provide practice to the students on summing up the square of each element.

Question 5

[15]

Define a ***class*** to accept a String and Print if it is a ***Super string*** or not. A ***String is Super*** if the ***number of uppercase letters*** are equal to the ***number of lower case letters***.

[Use Character & String methods only]

Example : "COmmITmeNt"

Number of Uppercase letters – 5

Number of Lowercase letters – 5

String is a Super String

MARKING SCHEME

```
import java.util.*;
class superstring
{
   String s;
   Scanner ob = new Scanner (System.in);
   void check()
   {
      System.out.println("Enter a string");
      s=ob.next();
      int i, l =s.length(),uc=0,lc=0; char ch;
      for(i=0;i<l;i++)
      {
        ch=s.charAt(i);
        if(Character.isUpperCase(ch))
        uc++;
         if(Character.isLowerCase(ch))
         lc++;
      }
      if(uc==lc)
      System.out.println("Super String");
      else
      System.out.println("Not a Super String");
   }
}
```

Comments of Examiners

Some of the common errors made by the candidates were:

- Did not use the character methods.
- Initialization of number of uppercase characters and number of lower-case characters was done inside the loop.
- String was initialized instead of accepting.
- Missed length ( ) .
- Used if(ch>='A' && ch<='Z") instead of character class method.
- Not printed the messages as 'Super String' or 'Not super String'.
- Wrote if(uc=lc) instead of if (uc== lc) .

Suggestions for teachers

- Instruct the students to read the question paper carefully and answer according to the requirements of the question.
- Encourage the students to make use of the wrapper class and string methods wherever possible.
- Guide the students to initialize the counter variables before the loop.
- Provide sufficient practice of questions which involve the wrapper class and string methods.
- Clearly highlight that when the question specifies print the message, students need to print the message.

Question 6

[15]

Define a ***class*** to initialise the following data in an array.

Search for a given character input by the user, using the ***Binary Search technique***.

Print ***"Search Successful"*** if the character is found otherwise print ***"Search is not Successful"***.

'A', 'H', 'N', 'P', 'S', 'U', 'W', 'Y', 'Z', 'b', 'd'

MARKING SCHEME

```
import java.util.*;
class search
{
   char ch[]={'A','H','N','P','S','U','W','Y','Z','b','d'};
   char n;
   Scanner ob = new Scanner(System.in);
   void search()
   {
      System.out.println("Enter character to be searched");
      n=ob.next().charAt(0);
      int f=0,l=ch.length,m;
      while(f<l)/while (f=l)
      {
         m=(l+f)/2;
         if(n==ch[m])
         {
           System.out.println("Search Successful");
           System.exit(0);
         }
         if(n>ch[m])
         f=m+1;
         if(n<ch[m])
         l=m-1;
      }
      System.out.println("Search is not successful");
   }
}
```

Comments of Examiners

Common observations were:

- Some candidates got confused as the array given was a character array.
- Some candidates wrote the program using linear search.
- Instead of initializing the array as specified, some candidates accepted the elements into the array double.
- Search element was not accepted.
- Integers were initialized / accepted instead of characters,
- length was written as length().
- Given while condition as (last<first).
- Wrongly placed statement.
- Missing the mid-value calculation.
- Checking for char using equals() instead of ==.
- Both if conditions were wrongly written.
- (a[mid]<search) was written as mid<search.
- First and last updating done incorrectly.
- Not used flag or System.exit().
- Printing both the messages was done inside loop itself, if search element was found.

Suggestions for teachers

- Explain the concept of sorting and searching programs by using integer arrays.
- Provide sufficient practice to the students of these programs with the different data types as int, char, String(only linear search).
- Ask students to initialize the values as given in the question.
- Explain the set of statements in binary search in detail.
- Encourage the students to try the binary search using the different data types initializing the array in a sorted format.
- Instruct students not to sort the array again.
- Lay stress on the usage of System. exit(0) when the search value is found, which is easier compared to the flag concept.

Question 7

Define a ***class*** to overload the method ***print( )*** as follows: **[15]**

void print ( ) – To print the given format using nested loops.

@#@#@

@#@#@

@#@#@

@#@#@

double print(double *a*, double *b*) – To display the sum of numbers between ***a*** and ***b*** with difference of 0.5.

e.g. if $a = 1.0$, $b = 4.0$

output is: 1.0 + 1.5 + 2.0 + 2.5 + ……. + 4.0

int print(char ch1, char ch2) – compare the two characters and return the ASCII code of the largest character.

MARKING SCHEME

```
class    overload
{       void  print ( )
 {      for( int i=1;i<=4;i++)
   {   for(int j=1;j<=5;j++)
      {         if( j%2==0)          ]
         System.out.print("#");
         else
         System.out.print("@");
       }
      System.out.println();
  }
}
double print( double a , double b)
{   double s=0.0;
  for( ;a<=b ;a+=0.5)
  {     s+=a;
  }
   return s;
}
 int  print( char ch1 , char ch2)
{       if(ch1>ch2)
    rerurn (int)ch1;
   else
    return (int)ch2;
}}
```

Comments of Examiners

Some common mistakes observed were:

- Wrote separate programs (polymorphism was not followed).
- The specifications given in the question were not followed.
- In pattern program, println() was used in both the loops.
- In first method , outer for loop and inner loop were reversed.
- In second & third method, double and System.out.print( ) were written together.
- for (double i=1.0 ; i<=b ;i++) { s+=a+0.5;}, some candidates were unable to construct for loop with a double value.
- System.out.print( ) was written instead of return statement.

Suggestions for teachers

- Clearly state the rules to be followed for using Polymorphism in the programs.
- Instruct the students to use the same method signature given in the question.
- Encourage the students to use System.out.println to display the format in the first part – nested loop format.
- Explain the concept of nested for loop and ask the students to write code for a variety of simple patterns.
- Provide rigorous practice to the students to write the code for the summation series involving integers and for the double values.
- Instruct the students to make use of the wrapper class methods.

Question 8

[15]

Define a ***class*** to accept a number. Check if the sum of the ***largest digit*** and the ***smallest digit*** is an even number or an odd number. ***Print appropriate messages.***

Sample Input:	6425	3748
Largest digit:	6	8
Smallest digit:	2	3
Sample Output:	Sum is even	Sum is odd

MARKING SCHEME

```
import  java.util.*;
class   check
{       void  main( )
   {  Scanner  sc=new Scanner(System.in);
      System.out.print("enter number");
      int  n=sc.nextInt();
      int  small=n%10 ;
      int large=n%10;
       while(n>0)
       {       int  rem=n%10;
         if(rem>large)
         large=rem;
       if(rem<small)
```

```
        small=rem;
       n=n/10;
     }
   if((small+large)%2==0)
   System.out.print("Sum is even");
    else
   System.out.print("Sum is odd");
} }
```

Comments of Examiners

- Many candidates were unclear about the extraction of digits and the manipulation of the digits of a number.
- Some of the candidates found the sum inside the loop.
- Some candidates wrote the incorrect while loop and used the wrong brackets.
- A few candidates did wrong initialization of max and min, while some other missed the update n/=10.
- A few candidates used the incorrect if condition sign to find max and min.

Suggestions for teachers

- Clarify the concept of extracting and manipulating the digit.
- Clearly explain to the students to manipulate the digit as per the specification mentioned in the question.
- Provide practice of the programs based on extraction, manipulation of digits of the number

GENERAL COMMENTS

Topics found difficult by candidates	• Package names. • Array initialization/accessing the elements of a 2D array. • Assertion and Reasoning. • Method overloading. • Nested loops.
Concepts in which candidates got confused	• One dimensional and two-dimensional array declaration and working and index positions. • Creation of object. • Constructor overloading. • Exit and entry-controlled loop. • Character wrapper class usage. • Public /private/protected. • Types of errors.

Suggestions for candidates

- Make your own notes on every topic and practice the concepts on a regular basis.
- Avoid selective study.
- Note down every minute detail of each program.
- Write the list of variables or prepare a variable description table for every program.
- Understand the concept by doing a dry run of the program with different inputs.
- Practice writing programs on paper before executing them on the system.
- Revise the subject at least twice a week.
- Be thorough with all wrapper class methods and string methods.
- For Section A, candidates need to be thorough with all the basic concepts.
- Make effective use of the fifteen minutes of reading time.
- Choose only one option while answering multiple-choice questions.
- Show the working steps for output-based questions.
- For any program-based question, understand the exact requirement of the question.
- Use the same variable names and method names as given in the question.
- Follow the specifications given in the question for method overloading.
- Write all answers in legible handwriting.
- Practice questions from previous years' board papers, specimen papers, and competency-based questions.